



O'Neill School of Public and Environmental Affairs

Rent Capture or Regional Forces? Fiscal Interaction in Metropolitan Areas

Steven Craig, Oscar Galvez-Soriano, Annie Yu-Hsin Hsu, Janet Kohlhase

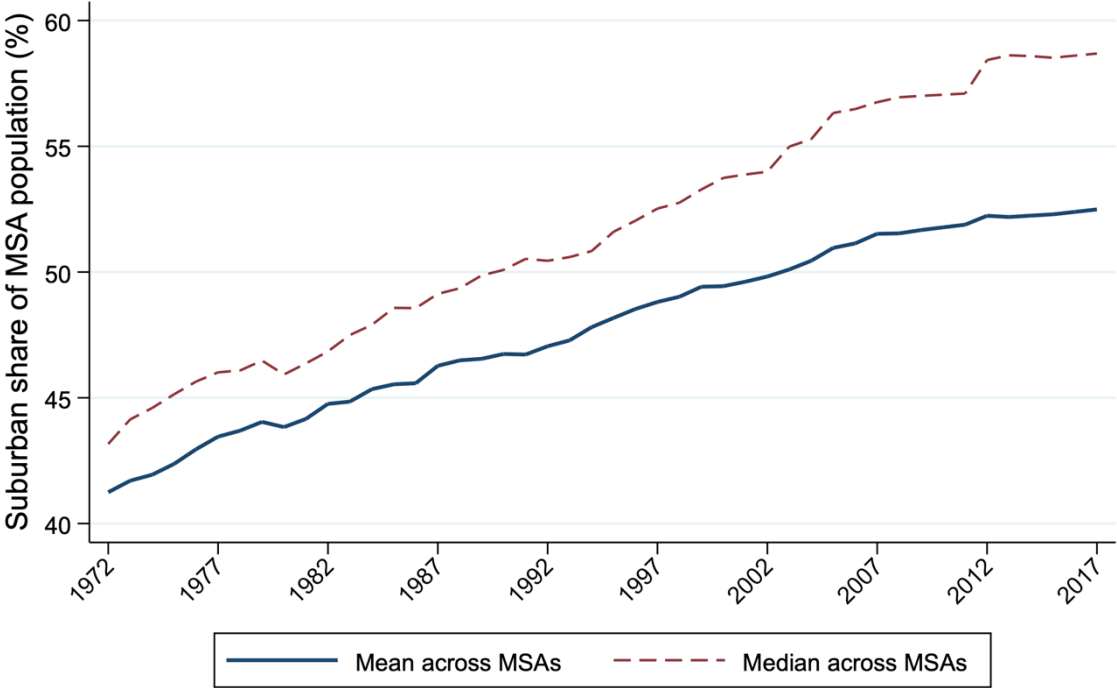
ABFM Conference · September 24, 2026

SECTION 1

Introduction

Metropolitan fiscal restructuring in the long run:

Suburban Population Share in Top-Quartile U.S. Metropolitan Areas
MSAs ranked by 1990 Census population, 1972-2017



- Suburban population shares have increased substantially
- Suburbanization reallocates tax bases across jurisdictions

Notes: The estimation sample consists of 52 MSAs located in states with more than one sample MSA.



Research question:

What drives metropolitan fiscal co-movement?

1. Strategic interaction:

- Central-city budgets respond to suburban fiscal expansion through strategic, political, or service-demand channels.

2. Common regional shocks:

- Cities and suburbs share labor markets, housing demand, and state institutions.

3. Contribution:

- Compare central-city tax and expenditure responses.
- Assess sensitivity to regional controls.
- Trace fiscal adjustment over multiple horizons.



Conceptual Framework: Rent Capture and Fiscal Fundamentals

1. Rent extraction mechanism

- Besley and Case (1995): political agency framework
- Some incumbents may set taxes above service cost
- Empirical implication: revenue growth without proportional service expansion

2. Why central cities may have fiscal slack?

- Legacy infrastructure and rigid expenditure commitments
- Broader service mandates and relatively immobile tax bases



Potential for asymmetric fiscal response to suburban expansion



Competing Mechanism:

Strategic interaction and common regional shocks

1. Brueckner (2001, 2003): Tax competition in spatial equilibrium
 - Strategic fiscal responses can generate city–suburb co-movement
 - Shared economic conditions can generate similar patterns
 - Common institutional constraints

2. Empirical prediction:
 - Division-by-year effects absorb broad regional shocks
 - Concern: how to separate directional fiscal response from shared regional economic forces



SECTION 2

Empirical Design

Data

1. Fiscal Data: U.S. Census Bureau's Annual Survey of State and Local Government Finances
 2. Population Data: U.S. Census Bureau's Population and Housing Unit Estimates Dataset
 3. Employment Data: Census Bureau's County Business Patterns
 4. Building Permits Survey (BPS): U.S. Census Bureau
 5. FHFA House Price Index: U.S. Federal Housing Finance Agency
 6. Migration Data: IRS SOI Migration data (ongoing)
 7. Quasi-experimental extension: New Jersey State digital library, Indiana Department of Local Government Finance (ongoing)
- Baseline period: 1972-2017 (subsample: 1990-2017)
 - Top Quartile US MSAs: 68 MSAs (preferred IV sample: 52 in the multiple-MSA-states)



Empirical Strategy

1. Estimating central city fiscal response to suburban fiscal expansion
 - Endogeneity addressed using an IV strategy
 - Instruments: suburban fiscal growth in other MSAs (out-of-state; within-state)
2. Testing mechanism via expenditure composition
 - Examine heterogeneous responses across spending categories:
 - Basic services: fire, police, libraries, parks, highways
 - Transfers: health, hospitals, welfare, housing
 - Other operations: corrections, judicial/legal, etc.



Summary Statistics:

Central cities and their suburbs, 1972–2017

Variable	All MSAs: City	All MSAs: Suburbs	City exceeds suburbs (%)	Multiple-MSA states: City	Multiple-MSA states: Suburbs	City exceeds suburbs (%)
Total revenue	2,094.9 (1,315.9)	1,076.8 (698.3)	94.5	2,147.9 (1,385.2)	1,114.4 (721.2)	92.7
Total taxes	688.4 (457.0)	384.8 (234.9)	78.9	708.4 (478.0)	397.1 (211.0)	78.4
Current operations	1,415.9 (971.7)	798.9 (574.4)	77.2	1,477.4 (1,028.3)	834.6 (604.7)	77.0
Basic expenditure	444.8 (153.7)	284.0 (121.0)	56.7	441.1 (154.1)	282.8 (123.8)	55.9
Transfer expenditure	177.6 (339.6)	41.5 (77.2)	327.9	186.5 (375.0)	45.5 (85.6)	309.9
Other expenditure	539.9 (508.9)	285.3 (288.3)	89.2	547.5 (517.9)	288.1 (279.0)	90.0
Observations	3,128	3,128		2,392	2,392	
MSAs	68	68		52	52	



Main Specification

$$\Delta y_{it}^{CC} = \beta \widehat{\Delta y_{it}^{Sub}} + \mu_i + \tau_t + \varepsilon_{it}$$
$$\Delta y_{it}^{CC} = \beta \widehat{\Delta y_{it}^{Sub}} + \mu_i + \lambda_{d(i),t} + \varepsilon_{it}$$

- μ_i : MSA fixed effects
- τ_t : common year effects
- $\lambda_{d(i),t}$: Census division x year effects that absorb region –specific annual shocks
- β : elasticity of central city fiscal growth with respect to instrumented suburban fiscal growth

The IVs use fiscal changes outside the focal MSA

$$\Delta y_{it}^{Sub} = \pi_{in} Z_{it}^{in} + \pi_{out} Z_{it}^{out} + \mu_i + \lambda_{d(i),t} + u_{it}$$

- In-state IV: suburban fiscal growth in other MSAs in the same state
- Out-of-state IV: suburban fiscal growth in other MSAs outside the focal state, constructed leave-one-out
- Relevance: external suburban growth predicts local suburban growth
- Exclusion assumption: conditional on controls, the instrument affects central-city outcomes through local suburban fiscal growth



The IVs compare the focal MSA with external suburban fiscal changes

Instrument geography: Houston example



$$Z_{it}^{in} = \frac{1}{N_{s(i)} - 1} \sum_{j \neq i, s(j)=s(i)} \Delta y_{jt}^{Sub}$$

$$Z_{it}^{out} = \frac{1}{N - N_{s(i)}} \sum_{j: s(j) \neq s(i)} \Delta y_{jt}^{Sub}$$

- Identification uses external fiscal co-movement after MSA and regional-year controls; strength and pre-outcome timing provide observable diagnostics

Out-of-state variation carries the informative first stage

Included IVs	Regional controls	KP F	Interpretation
Both IVs	Division x Year	21.22 – 42.37	Adequate combined first stage
Out-state only	Division x Year	32.12 - 46.05	Strong
In-state only	Division x Year	0.01 - 11.46	Unstable

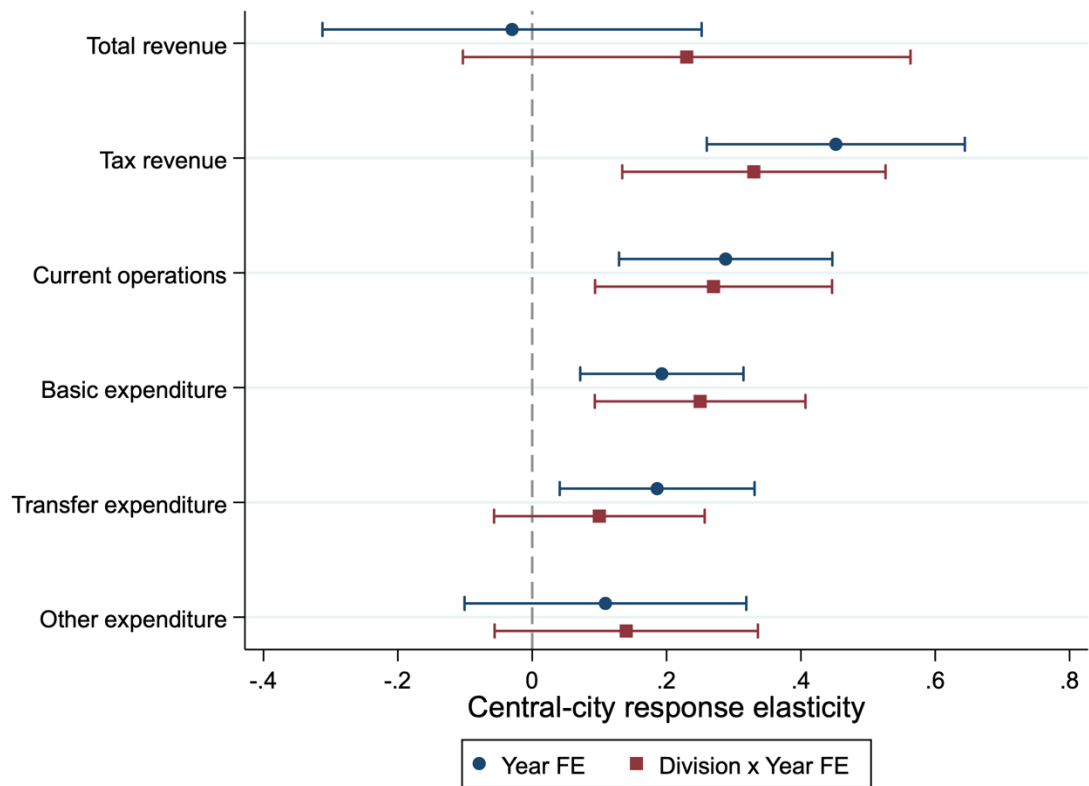
- The preferred diagnostic specification uses the strong out-of-state instrument alone.



SECTION 3

Results

Division-by-year effects attenuate the benchmark response



- Tax responses exceed basic-expenditure responses in point estimates.
- Moving from Year FE to Division-by-year FE attenuates the estimates, but the tax response remains economically meaningful.

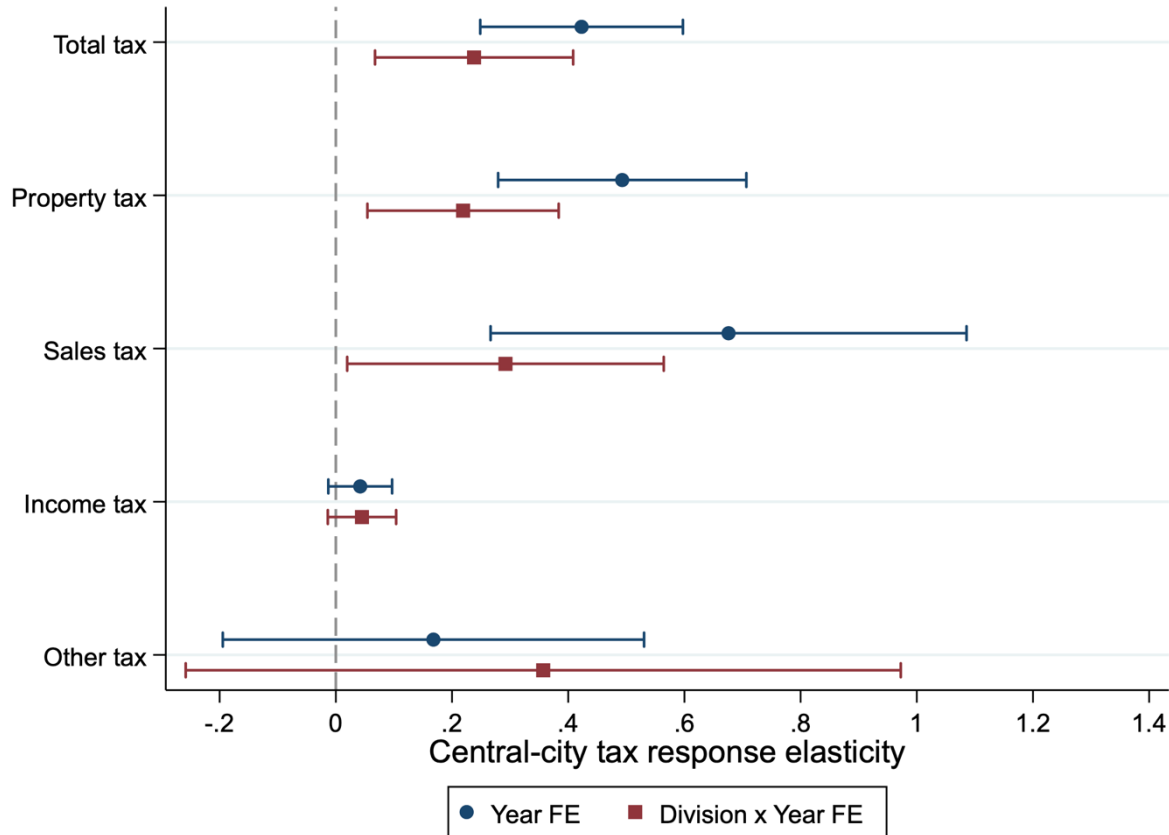
Opening the “regional forces” black box

Sample and specification	Tax response (SE)	BE response (SE)	N
MSA FE + Year FE : baseline specification	0.452*** (0.098)	0.193*** (0.062)	2,340
MSA FE + Division × Year FE	0.33** (0.10)	0.25*** (0.08)	2,340
MSA FE + Year FE + MSA- specific trends	0.465*** (0.100)	0.184*** (0.061)	2,340
Labor sample : baseline specification	0.455*** (0.118)	0.173** (0.075)	1,300
+ labor-demand shock	0.455*** (0.119)	0.173** (0.073)	1,300
with Division × Year FE	0.301*** (0.110)	0.271** (0.107)	1,300
BPS/HPI sample : baseline specification	0.423*** (0.142)	0.161* (0.0854)	943
+ building permits	0.418*** (0.144)	0.160* (0.0852)	943
+ HPI growth	0.425*** (0.145)	0.149* (0.0800)	943
+ building permits + HPI growth	0.424*** (0.145)	0.161** (0.0785)	943
with Division × Year FE	0.244** (0.120)	0.343*** (0.110)	943

- Labor and housing controls produce little change within matched samples.
- Division-by-year effects have a larger impact on the Tax–BE comparison.

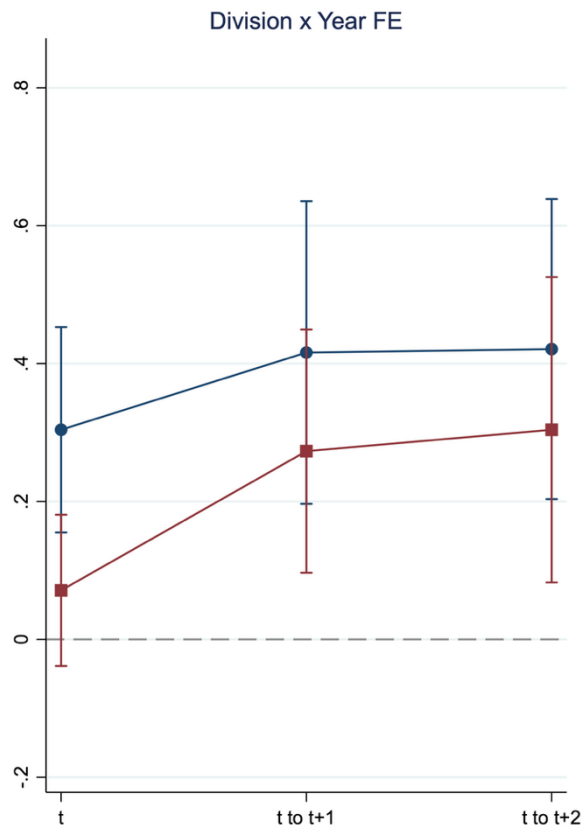
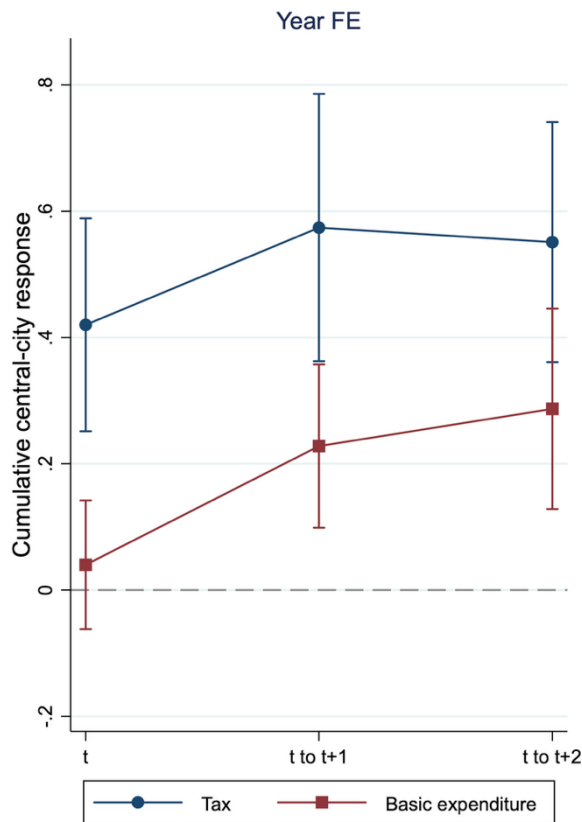


The tax response is concentrated in property and sales taxes



- Positive responses in property and sales taxes.
- Little evidence of an income-tax response.

Division-by-year effects narrow the cumulative Tax–BE gap



- The tax response extends beyond the contemporaneous period.
- The Tax–BE gap narrows at cumulative horizons under division-by-year effects.

SECTION 4

Identification and Robustness

No evidence of reverse timing

Pre-outcome	Year FE	Division $\times$ year FE
City Tax, t-2	-0.029 (0.069)	-0.013 (0.067)
City Tax, t-1	0.055 (0.060)	-0.016 (0.065)
City BE, t-2	0.003 (0.067)	-0.016 (0.073)
City BE, t-1	0.034 (0.050)	0.030 (0.068)
Joint p-value	0.862	0.959
N per equation	2,920	2,920

- No individual placebo is statistically significant.
- The four pre-outcomes are jointly insignificant under both FE structures.



Is the tax–expenditure gap statistically meaningful?

Fixed effects	Horizon	Tax (SE)	BE (SE)	Equality p
Year	t	0.420*** (0.086)	0.040 (0.052)	< .001
Year	through t+1	0.574*** (0.108)	0.228*** (0.066)	.001
Year	through t+2	0.551*** (0.097)	0.287*** (0.081)	.010
Division × year	t	0.304*** (0.076)	0.071 (0.056)	.005
Division × year	through t+1	0.416*** (0.112)	0.273*** (0.090)	.208
Division × year	through t+2	0.421*** (0.111)	0.304*** (0.113)	.297
N per equation		2,920	2,920	

- At the contemporaneous horizon, the difference is significant under both FE specifications.
- With division-by-year FE, equality is not rejected at cumulative horizons.



Fiscal persistence does not explain the cumulative estimates

Division × year FE outcome	No lag controls	+ Lagged city Tax and BE
Tax at t	0.304*** (0.076)	0.315*** (0.079)
BE at t	0.071 (0.056)	0.071 (0.056)
Cumulative Tax through t+1	0.416*** (0.112)	0.423*** (0.115)
Cumulative BE through t+1	0.273*** (0.090)	0.279*** (0.083)
Cumulative Tax through t+2	0.421*** (0.111)	0.430*** (0.114)
Cumulative BE through t+2	0.304*** (0.113)	0.314*** (0.104)
N per equation	2,920	2,851

- Adding lagged city tax and basic expenditure barely changes the coefficients.
- The cumulative pattern does not appear to be driven simply by fiscal persistence.



Interim Takeaway

1. The response is concentrated in property and sales taxes and persists beyond the contemporaneous period.
2. The tax–expenditure difference narrows under stronger regional controls and over longer horizons.

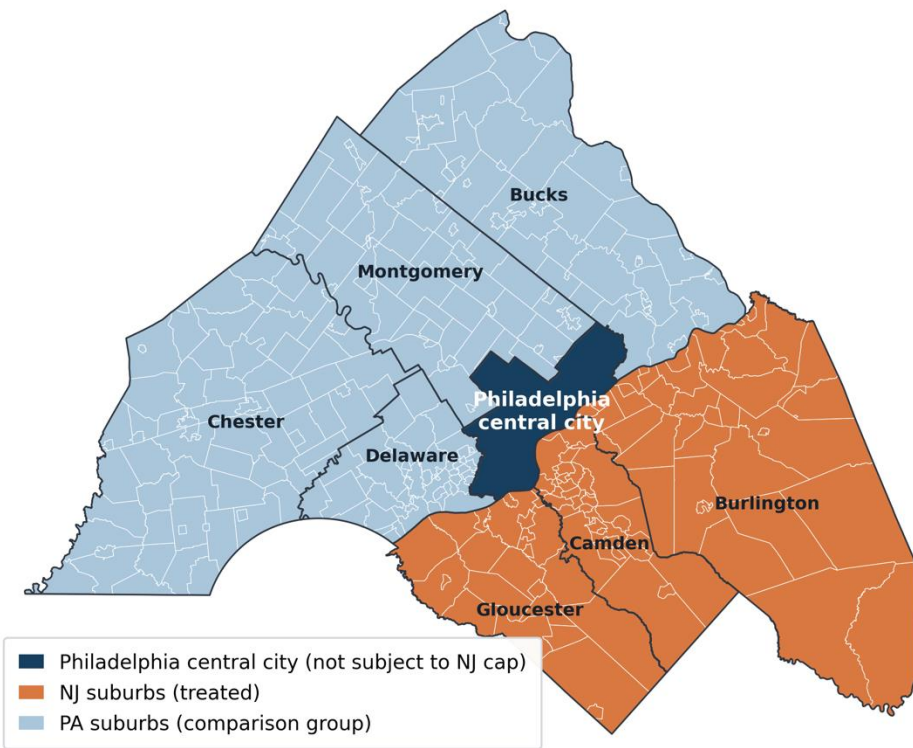


SECTION 5

Quasi-experimental validation (ongoing)

Ongoing design 1: New Jersey's Property-Tax Levy Caps

A state-border comparison within a common metropolitan economy



- **2007 law:** 4% levy-growth cap, with statutory exceptions.
- **2010 law:** 2% cap, applicable to calendar-year budgets beginning in 2011.
- **Setting:** NJ and PA suburbs within the Philadelphia metropolitan area.

Ongoing design 1: New Jersey's Property-Tax Levy Caps

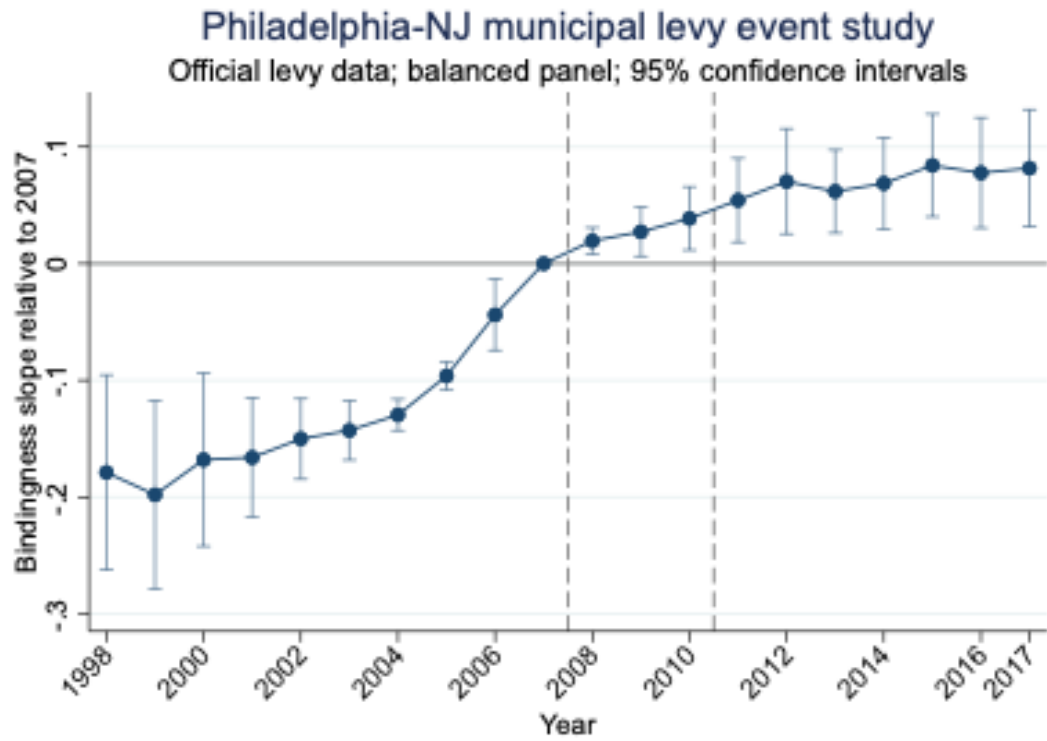
$$E_m = NJShare_{m,pre} \times StatutoryBindingness_{m,pre}$$

$$Z_{mt}^{(r)} = E_m \times Post_t^{(r)}, \quad r \in \{4\%, 2\%\}$$

- $NJShare_{m,pre}$: pre-reform share of suburban fiscal activity on the NJ side
- $StatutoryBindingness_{m,pre}$: pre-reform exposure to the statutory levy constraint
- $Post_t^{(r)}$: indicator for the cap-reform period
- **Current status:** annual levy data collected; historical cap worksheets remain incomplete.
- **Next step:** construct municipality-specific statutory bindingness net of exemptions, waivers, and cap banking.



Preliminary levy-exposure diagnostic: Philadelphia-Area NJ Municipalities

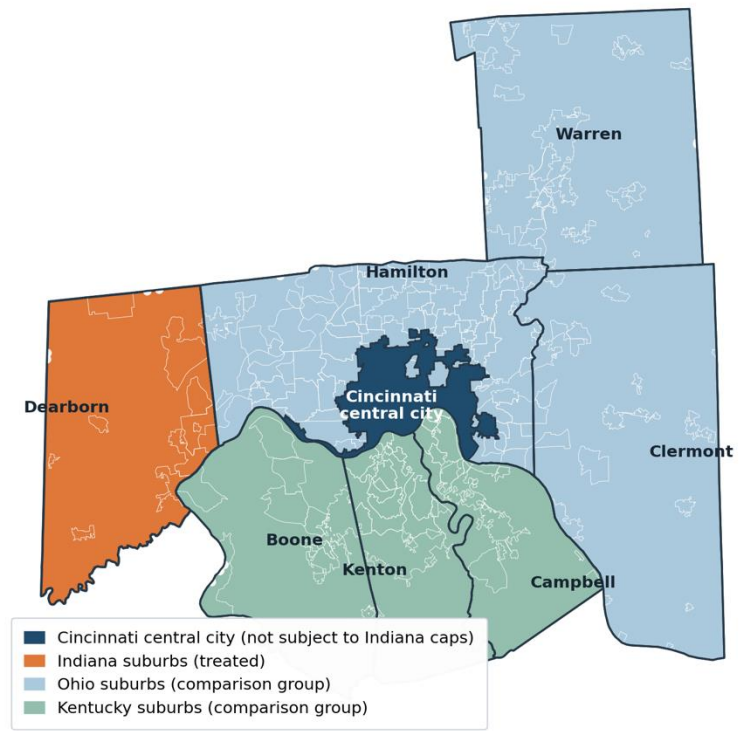


- 1. Levy paths differ by the exposure measure.
- 2. Substantial pre-reform trends require further investigation.



Ongoing design 2: Indiana Property-Tax Circuit-Breaker Caps

A state-border comparison within a common metropolitan economy



- 1. Phased implementation culminated in the full caps in 2010.
- 2. Indiana-side suburbs were directly exposed; Cincinnati and the Ohio/Kentucky sides were outside Indiana's statutory regime.

Cincinnati PMSA: Indiana-side suburbs in Dearborn County are exposed to the state caps; Ohio- and Kentucky-side suburbs provide distinct within-MSA comparisons.
1990 PMSA definition; 2024 place boundaries shown for geographic context.

Ongoing design 2: Indiana Property-Tax Caps

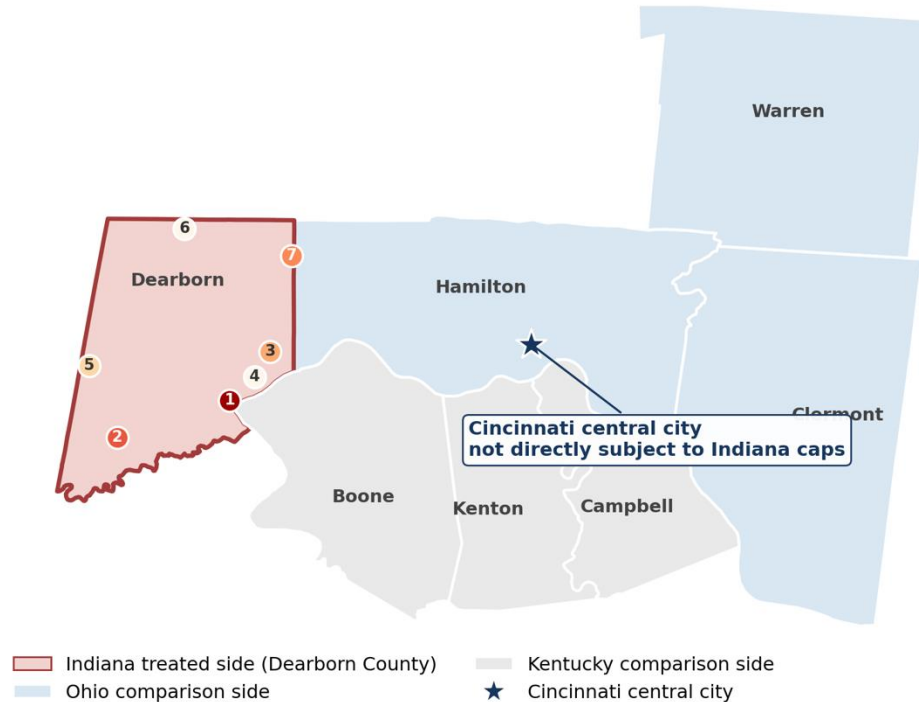
1. Circuit-breaker credits reduce revenues available to taxing units.
2. Aggregate exposure: 2.5% in 2010, 1.9% in 2011, and 3.9% in 2013.
3. Next step: link unit-level exposure to the border panel.
4. Measurement note:
 - 2010–2011: projections; 2013: reported credits.
5. $Z_{mt} = \text{INShare}_{m,\text{pre}} \times \text{CapBindingness}_{m,\text{pre}} \times \text{CapIntensity}_t$

$$\text{CapIntensity}_t = \begin{cases} 0, & t < 2008, \\ \text{Partial}, & t = 2008, 2009, \\ 1, & t \geq 2010. \end{cases}$$



Indiana property-tax-cap exposure within the Cincinnati MSA

Fixed 1990 PMSA counties; 2011 DLGF unit report



Dearborn County cities/towns

2011 credit / certified levy

1	Aurora	7.4%
2	Dillsboro	5.2%
3	Greendale	3.4%
4	Lawrenceburg	0.0%
5	Moores Hill	1.8%
6	St. Leon	0.0%
7	West Harrison	4.1%

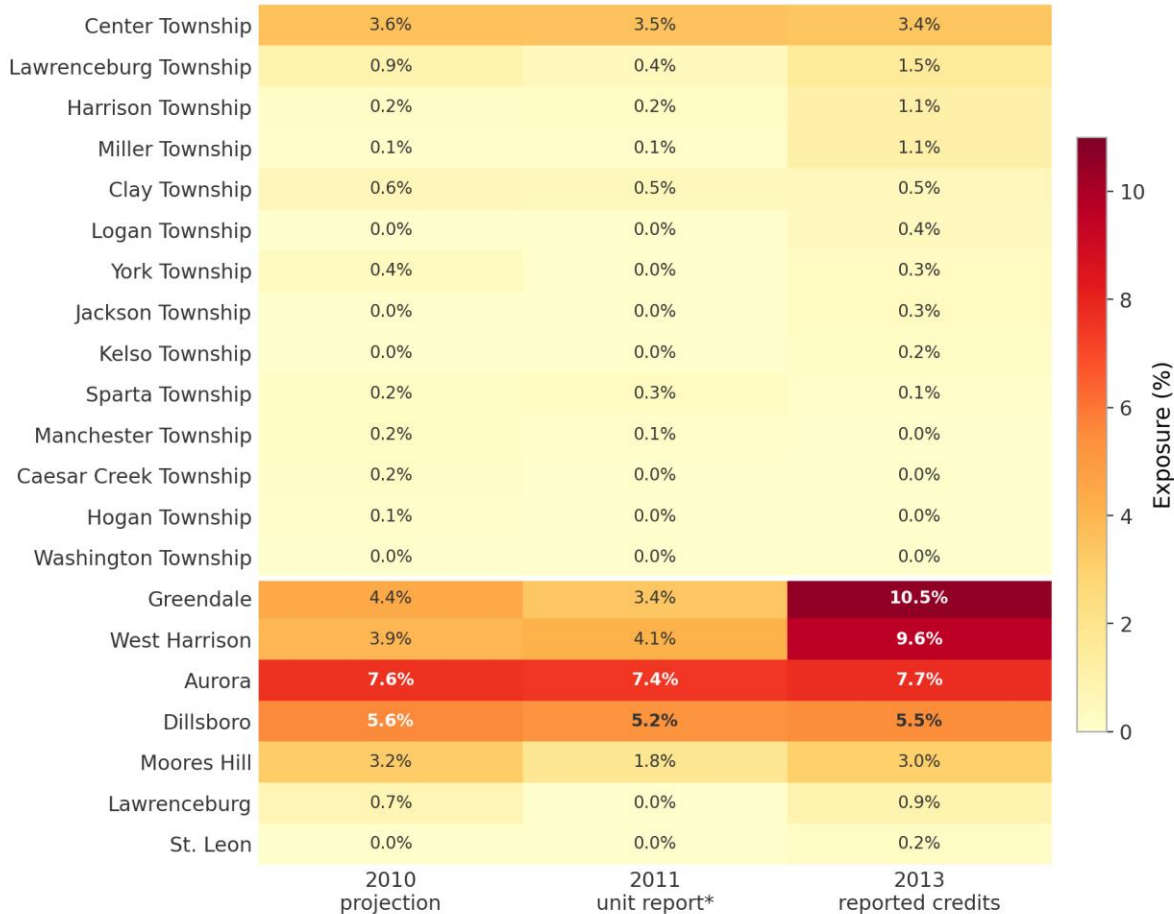
**Treatment intensity varies substantially
within the Indiana side of the MSA.**

Notes: Markers show Dearborn County cities/towns; townships are excluded from the map because their jurisdictions overlap incorporated places. The 2011 DLGF report is abstract-based and described in its footnote as a projection. Sources: Indiana Department of Local Government Finance; U.S. Census Bureau cartographic boundaries (2024).



Circuit-breaker exposure by local government and year

Credit as a percentage of levy; Dearborn County



Conclusion

1. Central-city tax responses are positive, especially for property and sales taxes.
2. The Tax–BE gap narrows at longer horizons and under division-by-year effects.
3. Evidence for persistent rent capture remains limited; institutional designs are ongoing.



Appendix / Q&A

Appendix: Benchmark IV — Year Fixed Effects

First stage and 2SLS estimates using the out-of-state instrument only

	Total revenue	Tax revenue	Current operations	Basic expenditure	Transfer expenditure	Other expenditure
Panel A. First stage						
Out-of-state IV	-10.625*** (1.905)	-14.419*** (1.691)	-10.970*** (1.792)	-14.006*** (1.886)	-9.413*** (1.701)	-15.836*** (1.958)
N	2,340	2,340	2,340	2,340	2,337	2,340
Panel B. Structural equation (2SLS)						
Suburban fiscal growth	-0.030 (0.144)	0.452*** (0.098)	0.288*** (0.081)	0.193** (0.062)	0.186* (0.074)	0.109 (0.107)
Kleibergen–Paap F	31.13	72.70	37.49	55.18	30.60	65.43

Notes: Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01. Preferred instrument: leave-one-out out-of-state suburban fiscal growth.



Appendix: Benchmark IV — Division × Year Fixed Effects

First stage and 2SLS estimates using the out-of-state instrument only

	Total revenue	Tax revenue	Current operations	Basic expenditure	Transfer expenditure	Other expenditure
Panel A. First stage						
Out-of-state IV	-12.64*** (2.12)	-17.44*** (2.63)	-12.88*** (1.90)	-16.20*** (2.50)	-8.82*** (1.55)	-14.97*** (2.33)
N	2,340	2,340	2,340	2,340	2,337	2,340
Panel B. Structural equation (2SLS)						
Suburban fiscal growth	0.23 (0.17)	0.33** (0.10)	0.27** (0.09)	0.25** (0.08)	0.10 (0.08)	0.14 (0.10)
Kleibergen–Paap F	35.54	43.83	46.05	42.11	32.12	41.35

Notes: Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01. Preferred instrument: leave-one-out out-of-state suburban fiscal growth.



Instrument comparison

Kleibergen-Paap first-stage F-statistics across instrument sets

FE structure	Instrument set	TR	Tax	CO	BE	TranE	OE
Year FE	Both IVs	17.03	30.09	22.82	26.66	14.25	24.14
	Out-state only	31.13	72.70	37.49	55.18	30.60	65.43
	In-state only	9.62	21.17	8.53	10.69	0.40	16.01
Division × year FE	Both IVs	25.58	24.06	42.37	38.12	21.22	24.99
	Out-state only	35.54	43.83	46.05	42.11	32.12	41.35
	In-state only	10.30	0.01	4.71	4.54	11.46	4.60

Preferred IV: Out-of-state suburban fiscal variation provides both stronger conditional relevance and cleaner separation from focal-state fiscal policy shocks.



Diagnostic: Why Prefer the Out-of-State IV?

Outcome	Raw In	Raw Out	Year FE In	Year FE Out	Div×Year In	Div×Year Out
TR	.252	.286	.091	−.459	−.178	−.406
TT	.424	.298	.296	−.563	−.005	−.510
CO	.301	.344	.125	−.474	−.137	−.431
BE	.318	.335	.152	−.501	−.096	−.440
TranE	.078	.101	.022	−.275	−.225	−.238
OE	.290	.236	.186	−.532	−.118	−.469

- The in-state instrument is not inherently uncorrelated with suburban fiscal changes.
- Rather, much of its predictive content is associated with common geographic/time variation; once division-by-year shocks are absorbed, very little independent variation remains for tax revenue.



Tax decomposition regression table

Fixed effects	Total tax	Property	Sales	Income	Other	KP F / N
Year FE	0.423*** (0.089)	0.493*** (0.109)	0.676*** (0.209)	0.042 (0.028)	0.168 (0.185)	30.09 / 2,340
Division × Year FE	0.238*** (0.087)	0.219*** (0.084)	0.292** (0.139)	0.045 (0.030)	0.357 (0.314)	18.51 / 2,340

Property and sales-tax components carry most of the total-tax response; income-tax responses are small and statistically imprecise.

Notes: Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.



Regional controls and first-stage strength

Sample and specification	Tax response (SE)	Tax KP F	BE response (SE)	BE KP F	N
MSA FE + Year FE : baseline specification	0.452*** (0.098)	72.70	0.193*** (0.062)	55.18	2,340
MSA FE + Division × Year FE	0.33** (0.10)	43.83	0.25*** (0.08)	42.11	2,340
MSA FE + Year FE + MSA-specific trends	0.465*** (0.100)	71.97	0.184*** (0.061)	54.97	2,340
Labor sample : baseline specification	0.455*** (0.118)	62.30	0.173** (0.075)	46.06	1,300
+ labor-demand shock	0.455*** (0.119)	62.26	0.173** (0.073)	46.74	1,300
with Division × Year FE	0.301*** (0.110)	27.03	0.271** (0.107)	28.83	1,300
BPS/HPI sample : baseline specification	0.423*** (0.142)	48.29	0.161* (0.0854)	46.55	943
+building permits	0.418*** (0.144)	46.57	0.160* (0.0852)	46.20	943
+HPI growth	0.425*** (0.145)	46.24	0.149* (0.0800)	46.55	943
+building permits +HPI growth	0.424*** (0.145)	45.93	0.161** (0.0785)	46.01	943
with Division × Year FE	0.244** (0.120)	26.55	0.343*** (0.110)	34.37	943



Robustness check:

Baseline characteristics interacted with year effects

Baseline exposure × Year	Tax response (SE)	Tax KP F	BE response (SE)	BE KP F	N
Suburban population share	0.460*** (0.099)	77.57	0.220*** (0.062)	52.84	2,340
Fragmentation: Numbers of suburban govts	0.446*** (0.098)	75.42	0.193*** (0.062)	55.12	2,340
Fiscal dependence	0.477*** (0.104)	74.58	0.204*** (0.061)	68.37	2,340
Spatial structure: average ring	0.442*** (0.097)	76.92	0.202*** (0.060)	62.21	2,340



Expanded-Sample Robustness: Out-of-State IV Using 68 MSAs

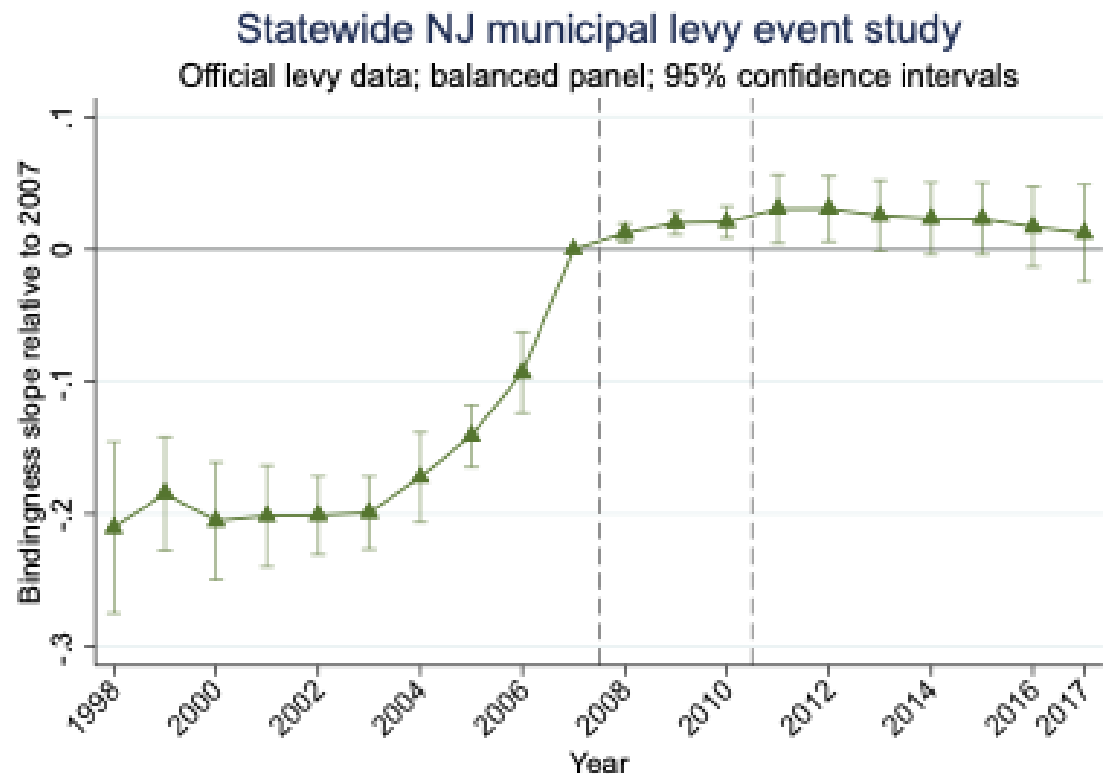
Outcome	Year FE	Division × Year FE
Total Revenue	0.060 (0.121)	0.17 (0.13)
Tax Revenue	0.436* (0.086)	0.33* (0.08)
Current Operations	0.248*** (0.067)	0.17* (0.07)
Basic Expenditure	0.150** (0.052)	0.17** (0.06)
Transfer Expenditure	0.190** (0.058)	0.11 (0.06)
Other Expenditure	0.091 (0.087)	0.06 (0.07)
MSAs	68	68
KP F-stat., range	31.8–84.4	30.5–72.7

- Expanded-sample estimates closely track the 52-MSA benchmark, indicating that the results are not driven by the original sample restriction required for the in-state IV.



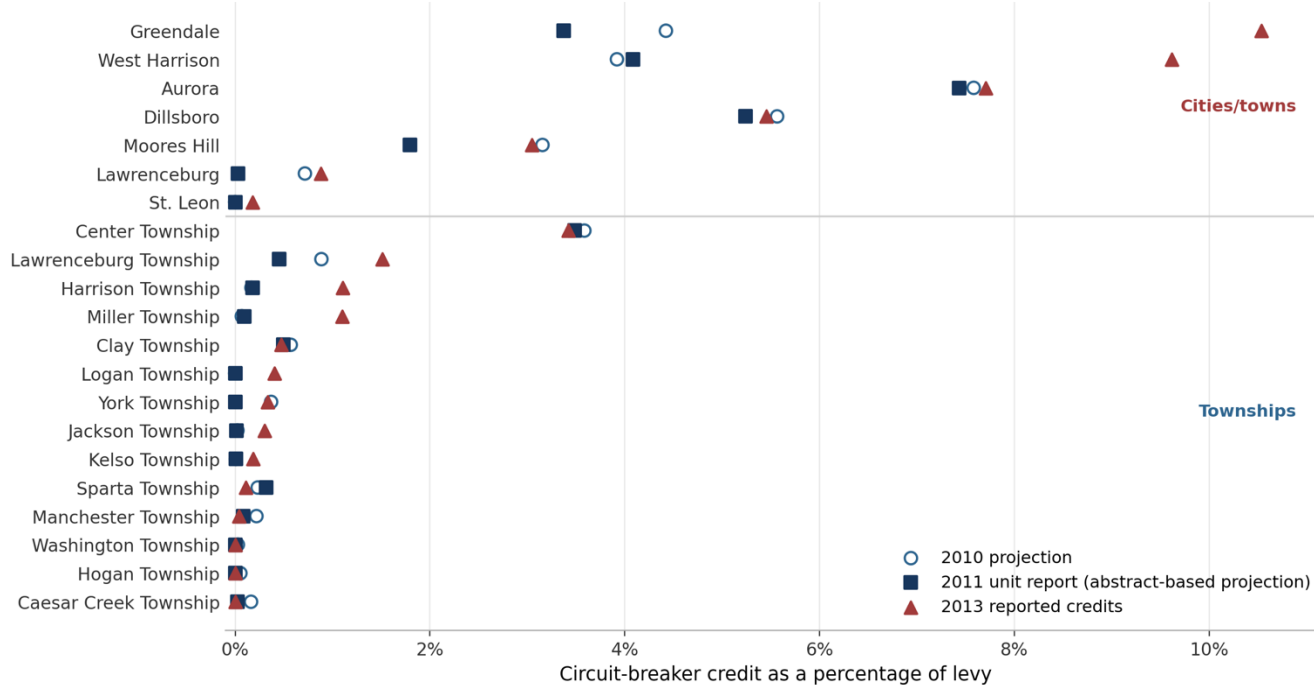
Appendix: Ongoing design 1: New Jersey's Property-Tax Levy Caps

NJ exposure-gradient diagnostic: statewide sample



Policy exposure differs across units and implementation years

Dearborn County general-purpose governments



Notes: Exposure uses certified levy in 2010 and 2011 and abstract levy in 2013. The 2010 and 2011 DLGF files are abstract-based projections; the 2013 file reports circuit-breaker credits by unit and fund. Comparisons across years should therefore be interpreted cautiously. Source: Indiana Department of Local Government Finance.

